

# Gravity sourced by records: a non-signalling semiclassical law, its predictions, and the limit on testing it

David T. Swanson   
(Dated: October 1, 2026)

I propose that gravity couples to a quantum system’s state conditioned on the physical records of its position that actually exist, whether or not they are ever read, rather than to its full wavefunction or to what an observer has measured. The law is stated in density-matrix language, provably does not signal, and has one closed form selected by requiring that independent records compose multiplicatively. It predicts no gravitational entanglement between two unrecorded masses, with no noise added to enforce that; and no self-gravity frequency shift for any mechanical oscillator above its motional ground state. I verify these properties by exact simulation, with controls, and then ask what a real experiment could see. On every platform considered, the records that suppress the predicted signal also broaden the spectral feature that would carry it, independent of the oscillator’s mass; this limit is specific to the present law, and a measurement-conditioned alternative evades it. A weaker bound applies to the whole class: any spectroscopic search for a conditioned second resonance needs an observation time of order  $\pi Q/\omega_0$ , a special case of a bound derived for measurement-conditioned Schrödinger–Newton gravity, and consistent with the run time reported by a recent torsion-balance experiment.

## I. INTRODUCTION

Whether gravity couples to quantum matter as a quantized field or through some semiclassical channel is open, and if the latter, the open question sharpens: semiclassical gravity needs a source, and *which state* of a quantum system that source is drawn from is itself a choice with physical consequences. Two answers are on the table. Unconditional Schrödinger–Newton gravity sources the field from a system’s full wavefunction, whatever branch it occupies and whatever records of it exist; the resulting nonlinear, nonlocal theory is not excluded by any experiment performed to date, but is argued on general grounds to permit superluminal signalling [1] and is in direct tension with the one experiment built to distinguish it from the ordinary, linear quantum expectation value [2]. Measurement-conditioned proposals instead source the field from what an experimenter’s own readout has resolved [3–5]; this avoids sourcing by an unmeasured superposition, but makes the source depend on a choice — what to measure, and when.

This paper takes a third option: condition the source not on the full wavefunction, and not on what has been measured, but on whatever physical *records* of a system’s position exist, independent of whether anyone ever reads them. A record is a copy — a correlation with the environment, a latch in an apparatus, a decohered pointer state — and its existence is a fact about the world, not about an observer’s choice. I show that the resulting law cannot signal (Sec. II), and derive its consequences for two canonical scenarios: two masses held in spatial superposition with no record of either taken (Sec. III), and a single oscillator’s resonance spectrum, where the *same* source rule makes a sharp, nearly parameter-free prediction depending only on whether the oscillator’s own thermal state has been reduced to its motional ground state.

Section IV verifies the law’s properties by exact simulation in a small model, with controls showing that the

simulation distinguishes the law from each alternative. Section V extends the law from two branches to a continuously decohered mechanical oscillator and evaluates it on three published platforms. The result is negative: the mechanism that makes the predicted signal small also makes the spectral feature carrying it broad, and the two do not separate. Section VI shows why this limit is specific to the present law, and Sec. VII identifies the weaker limit common to every proposal of this kind. Section VIII compares the law with other proposals, and Sec. IX lists the limitations of the analysis.

I present the law as a hypothesis with computable consequences, and identify what would test it.

## II. THE LAW

### A. A record, in density-matrix language

Let a quantum system occupy two distinguishable branches, labelled  $A$  and  $B$  — the two arms of an interferometer, the two positions of a levitated mass, the two outcomes a decay could leave behind. A *record* of which branch is a correlation with some other degree of freedom, a register, whose two conditional states  $|r_A\rangle$  and  $|r_B\rangle$  overlap at

$$\eta \equiv |\langle r_A | r_B \rangle| \in [0, 1]. \quad (1)$$

$\eta = 1$  is no record at all (the register carries no information about which branch occurred);  $\eta = 0$  is a perfect record. This is not a new quantity:  $\eta$  is exactly the system’s own decoherence factor, the off-diagonal element of its reduced density matrix in the branch basis, normalized to the branch populations. For an exactly balanced two-branch state,

$$\eta = \frac{|\rho_{AB}|}{\sqrt{\rho_{AA}\rho_{BB}}} = \sqrt{2\text{Tr}\rho^2 - 1}, \quad (2)$$

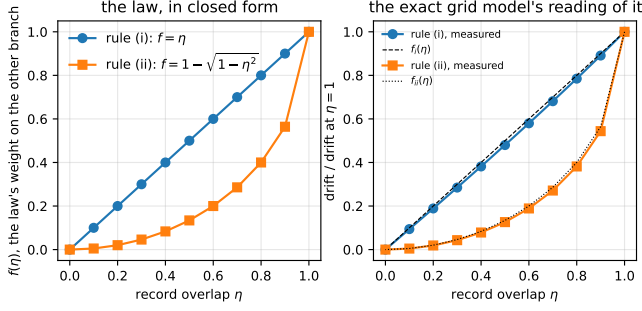


FIG. 1.  $f(\eta)$  for both closed forms, Eq. (5) (left), and an exact numerical model's reading of it (right): the measured drift between two branches, relative to its  $\eta = 1$  value, tracks  $f(\eta)$  to within a few per cent. At  $\eta = 0.5$ ,  $f_{(i)} = 0.5000$  against a measured ratio of 0.4800;  $f_{(ii)} = 0.1340$  against 0.1262. The two forms differ by a factor of 3.732 at  $\eta = 0.5$ .

the second equality following from  $\text{Tr } \rho^2 = (1 + \eta^2)/2$  for a two-branch state at equal weight. Equation (2)'s right-hand side is defined for *any* state, not only a balanced two-branch one, and this is what lets Sec. V extend the law to a real, continuously decohering oscillator with no branches in it at all.

The quantity  $\eta$ , and hence everything that follows, is a property of the system's own *reduced* state  $\rho$ , read in whatever basis locality fixes. It says nothing about whether the register has been read, by whom, or in what basis a distant party chooses to read it; Appendix B uses this to prove that the law does not signal.

### B. The source

The gravitational source is the record-conditioned density  $n_r$ , built to interpolate between two fixed points: with no record ( $\eta = 1$ ),  $n_r$  is the ordinary quantum-mechanical mean density,  $n_r = (n_A + n_B)/2$ ; with a perfect record ( $\eta = 0$ ),  $n_r$  is the localized branch alone,  $n_r = n_A$  (for the branch-*A* description). At intermediate  $\eta$ , two forms interpolate:

$$(i) \quad n_r^A = \eta \frac{n_A + n_B}{2} + (1 - \eta) n_A, \quad (3)$$

$$(ii) \quad n_r^A = (1 - P) n_A + P n_B, \quad P = \frac{1 - \sqrt{1 - \eta^2}}{2}. \quad (4)$$

Writing  $f(\eta)$  for the weight the law places on the branch a given description does *not* occupy, normalized to its no-record value of one half, the two forms are

$$f_{(i)}(\eta) = \eta, \quad f_{(ii)}(\eta) = 1 - \sqrt{1 - \eta^2}. \quad (5)$$

Both satisfy  $f(1) = 1$  and  $f(0) = 0$ , are continuous and monotonic on  $[0, 1]$ , and contain no free parameters. Figure 1 plots both, together with the drift between two branches they produce in an exact numerical model; the drift follows  $f(\eta)$  to within 4–6%, the difference coming from each branch's own wavepacket spreading.

### C. Properties, and the form adopted

*No signalling.* Both forms are functions of  $\eta$  alone, and  $\eta$  is a function of the system's own reduced state, Eq. (2). No local operation on the register — reading it, not reading it, applying any unitary to it whatsoever — can change the system's reduced state, so neither form of the law can be used to signal (Appendix B). Numerically, the prediction is unchanged to  $2.220e - 16$  under twenty random unitaries on a two-dimensional register and to  $2.220e - 16$  on a four-dimensional one, for both forms. A law sourced instead by the outcome actually read changes by 0.3402 of its no-record value under the same test.

*Composition.* Two independent records at overlaps  $\eta_1, \eta_2$  are read by the law as a single record at overlap  $\eta_1 \eta_2$ , exactly, for *either* form (to  $5.551e - 17$ ), because the law reads only the system's reduced state, on which two independent records act multiplicatively in their overlaps.

*Multiplicativity.* A stronger condition is that  $f$  itself be multiplicative,  $f(\eta_1)f(\eta_2) = f(\eta_1 \eta_2)$ . Form (i),  $f(\eta) = \eta$ , satisfies it exactly (error 0); form (ii) misses by up to 0.07163 in absolute terms. This is the only criterion among these checks that separates the two forms; no signalling holds for both. I take multiplicativity as the selection principle and adopt form (i),  $f(\eta) = \eta$ , as the law in what follows. The principle is a requirement imposed here, not a consequence of the law's construction (Sec. IX). Form (ii) is retained in the figures and tables as the alternative.

*No added noise.* Unlike a genuinely classical gravitational channel, which must add decoherence to the systems it couples as the price of not signalling [6], this law's mean-field dynamics is a *deterministic* function of the state, with no stochastic term. The law avoids signalling by conditioning on a basis-independent fact, how much record exists, rather than by adding noise to wash out a dependence on what is read.

## III. PREDICTIONS

*a. P1: gravity without entanglement, and without added noise.* Two masses, each held in a spatial superposition with no record of either taken, interact only through their respective mean fields under this law. Simulated exactly as two qubits, the state's negativity stays below  $1.000e - 10$ , zero to numerical precision, while a quantized two-body coupling applied to the same initial state reaches a concurrence of 0.5274 (Fig. 2). The masses still attract: each accumulates a relative phase from the other's mean field. What is absent is entanglement and, unlike a classical-channel model [6], which also predicts no entanglement but adds noise, any added noise. This is the discriminator the gravitationally-induced-entanglement proposals [7, 8] are built to test: if gravity mediates entanglement between two otherwise-unconnected quantum systems, its mediator must itself

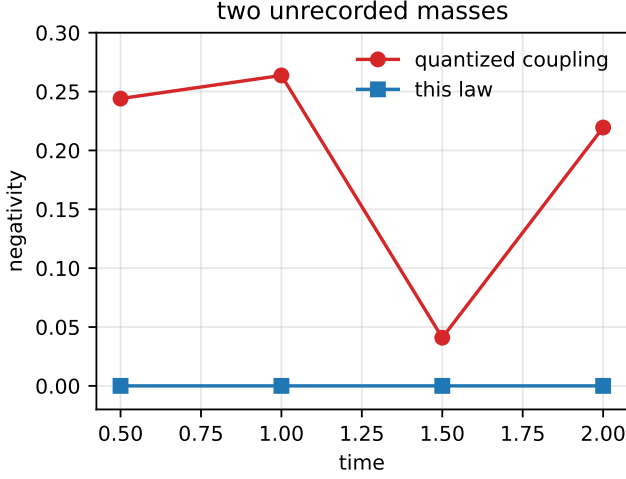


FIG. 2. Two unrecorded masses: negativity under this law’s mean-field form (flat, below  $1.000e-10$ ) against a quantized two-body coupling applied to the same initial state (reaching concurrence 0.5274), which serves as the control.

be quantum. Under this law, with no record taken, it would not. The simulation is mean-field rather than a full master equation (Sec. IX).

*b. P2: no self-gravity shift above the ground state.* For a single mechanical oscillator, the law’s weight on the unconditioned state,  $f(\eta)$ , is nonzero only where the oscillator’s own thermal occupation satisfies  $\bar{n} < 1/2$  — i.e., only in or near the motional ground state — because  $\eta_{\text{eff}} = \sqrt{2 \text{Tr} \rho^2 - 1}$  with  $\text{Tr} \rho^2 = 1/(2\bar{n} + 1)$ , and both factors are forced by Eq. (2) (Sec. V). For any oscillator with  $\bar{n} > 1/2$ , this includes all three platforms considered here, the resonance sits exactly at the bare mechanical frequency  $\omega_0$ , with *no* self-gravity shift at all. Unconditional Schrödinger–Newton, by contrast, shifts the resonance to  $\omega_q = \sqrt{\omega_0^2 + \omega_{\text{SN}}^2}$  regardless of the oscillator’s thermal state. Unconditional Schrödinger–Newton gravity is not excluded by any published experiment (Sec. VIII), so this is a live discriminator.

*c. P3: record dependence in the ground state.* Only for an oscillator cooled below its motional ground-state occupation threshold does the record weight  $f(\eta)$  depart from zero, and with it the signatures that distinguish an unread record from a read one, and a held record from an erased one. Section V shows these signatures are out of reach on every platform considered: the one platform pure enough to have  $f \neq 0$  is also the one where the predicted signal is smallest relative to the noise.

Table I collects the four scenarios across the laws compared.

#### IV. EXACT CHECKS

The law and its three comparison laws (unconditional Schrödinger–Newton, a measurement-conditioned alter-

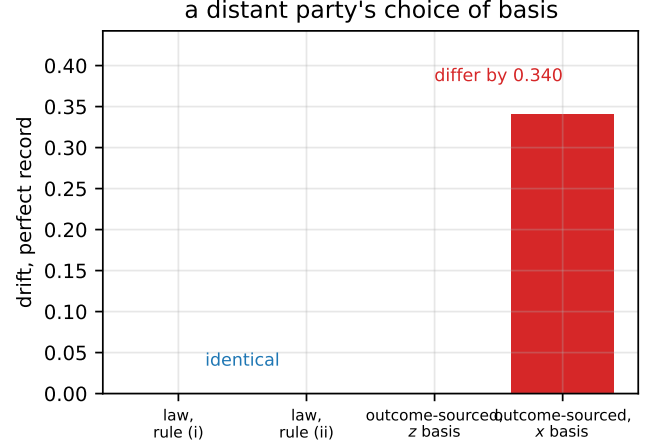


FIG. 3. A distant party’s choice of measurement basis on a perfect record: the law’s prediction (both forms) is identical in either basis; an outcome-sourced alternative’s differs by 0.3402 of its no-record value between them, and so signals.

native that sources by the outcome actually read, and standard quantum mechanics with no self-gravity term) are implemented in one shared code path, each differing only in what the conditioned density  $n_r$  is set to. Two models carry the checks: an exact split-step propagation of two branch wavefunctions, each in its own record-conditioned potential, on a few hundred grid points; and exact linear algebra on a register Hilbert space of at most a few hundred dimensions. Nothing is sampled or fitted.

Table II summarizes the nine tests. Three controls show that the simulation distinguishes the laws: unconditional Schrödinger–Newton is flat in  $\eta$ , as it must be; the quantized two-body coupling generates entanglement where the law’s mean-field form does not (Fig. 2); and the outcome-sourced alternative differs from the law when a record goes unread.

Figure 3 shows the no-signalling check directly.

For the conservation test (T9), the conserved quantity is the functional that generates the dynamics,

$$E = T_A + T_B + \frac{\alpha}{2} (\langle n_A K n_A \rangle + \langle n_B K n_B \rangle) + \beta \langle n_A K n_B \rangle, \quad (6)$$

with  $\alpha, \beta$  the law’s two coefficients. (Summing each branch’s potential against the other branch’s density instead double-counts the self-interaction.) The worst drift is  $2.980e-4$ , and it falls by a factor of 4.03 when the time step is quartered, as expected for first-order integrator error. Across a change of record the energy changes by 52.41%, because changing the record changes the Hamiltonian; testing conservation across such a change requires the full mass-plus-register system, which this reduced model does not include (Sec. IX).

TABLE I. Discriminators: this law (both forms) against quantized gravity, standard quantum mechanics, and the unconditional and measurement-conditioned Schrödinger–Newton alternatives, across four scenarios. Dashes mark combinations not modelled here; see the footnotes.

| scenario                               | this law (i)   | this law (ii)  | quantized grav. | standard QM    | uncond. SN     | meas.-cond. SN      |
|----------------------------------------|----------------|----------------|-----------------|----------------|----------------|---------------------|
| two unrecorded masses (negativity)     | 0 <sup>†</sup> | 0 <sup>†</sup> | 0.264           | 0 <sup>‡</sup> | — <sup>a</sup> | — <sup>a</sup>      |
| oscillator above ground state (shift?) | no             | no             | no <sup>b</sup> | no             | yes            | no                  |
| unread record in the ground state      | 0.4800         | 0.1262         | — <sup>c</sup>  | — <sup>c</sup> | 1.0000         | 1.0000 <sup>d</sup> |
| the same record, erased                | 1.0000         | 1.0000         | — <sup>c</sup>  | — <sup>c</sup> | 1.0000         | 1.0000 <sup>d</sup> |

<sup>†</sup>Exact two-qubit simulation, mean-field form (Fig. 2). <sup>‡</sup>Zero by construction: no coupling term exists between the masses. <sup>a</sup>Not modelled here: neither unconditional nor measurement-conditioned Schrödinger–Newton has a two-mass mechanism implemented here; the classical-channel answer is no entanglement with added noise [6]. <sup>b</sup>No two-body coupling partner exists in a single-oscillator spectrum; trivially identical to standard quantum mechanics in this scenario. <sup>c</sup>No register is implemented for these laws here. <sup>d</sup>Unread means nothing has yet been read, so the measurement-conditioned source has not conditioned on anything — identical to the no-record case here.

TABLE II. The nine tests and their results.

| test | what it checks                         | result                                |
|------|----------------------------------------|---------------------------------------|
| T1   | the law’s two endpoints                | pass (both rules)                     |
| T2   | the curve between the endpoints        | pass; continuous, monotonic           |
| T3   | composition; multiplicity of $f$       | exact to 5.6e-17; separates the rules |
| T4   | no signalling                          | pass to 2.22e-16 (both rules)         |
| T5   | an unread record counts fully          | pass                                  |
| T6   | erasure restores the source            | pass                                  |
| T7   | environment = deliberate register      | pass                                  |
| T8   | two unrecorded masses: no entanglement | negativity 1.8e-16 vs quantized 0.264 |
| T9   | no record configuration creates energy | worst drift 3.0e-04, integrator-order |

TABLE III. The three platforms, drawn from the published literature, with assumed values marked.

| platform     | $m$ [kg]  | $\omega_0/2\pi$ [Hz] | $T$ [K]   | $Q$       | ref. |
|--------------|-----------|----------------------|-----------|-----------|------|
| osmium-paul  | 1.661e-12 | 1.000e+01            | 1.000e-01 | 1.664e+09 | [9]  |
| torsion-mg   | 9.000e-05 | 2.600e-03            | 2.930e+02 | 5.000e+00 | [10] |
| levitated-gs | 3.368e-18 | 3.050e+05            | 1.218e-05 | 1.000e+08 | [11] |

cillates at  $\omega_c = \sqrt{\omega_0^2 + f\omega_{\text{SN}}^2}$ , and the fluctuation about it, which carries the conditional variance of whatever the law conditions on, sits at  $\omega_q = \sqrt{\omega_0^2 + \omega_{\text{SN}}^2}$ .

This construction extends the law beyond the two-branch states for which it is stated. Appendix C gives the filtering equations and an independent check showing that the conclusions below do not depend on the extension.

## V. UNDER REALISTIC CONDITIONS

### A. The adapter

The checks above use idealized two-branch states. A real test uses a massive mechanical oscillator, continuously decohered by its thermal environment and continuously read out by light. Extending the law to this setting takes three steps.

*How much record exists.* Equation (2)’s right-hand side,  $\eta_{\text{eff}} = \sqrt{2\text{Tr}\rho^2} - 1$ , is read off the oscillator’s *unconditioned* steady-state covariance — the state whose mixedness the environment’s records caused.

*What the source is.* The law’s own interpolation is applied with the unconditioned density playing the no-record role and the state *conditioned on every record that exists* (the continuous-measurement filtering solution, at unit efficiency on every decoherence channel) playing the perfect-record role. This reduces exactly to the two-branch law in the two-branch limit.

*What it does.* In the regime where the source’s spread is small against the material’s lattice localization length, the self-gravity term is harmonic, and the conditioned source produces *two* resonances: the conditional mean os-

### B. Three platforms

Table III lists the three platforms: a superconducting osmium microdisc in a Paul trap [9], the published proposal’s own choice of material and trap frequency; a milligram torsion balance [10], the lowest mechanical frequency measured to date; and a ground-state-cooled levitated nanosphere [11], the only platform of the three whose thermal occupation is below one half, and so the only one where  $f(\eta)$  is nonzero at all. This sphere is silica, for which no crystal-regime self-gravity constant is published; silicon’s is used instead, which overstates the effect and is therefore conservative for the negative conclusion below (Appendix D). Parameters not taken from their sources are marked as assumed and swept over many orders of magnitude.

### C. The result: a shape failure

On every platform, the law’s conditioned second resonance is broader than its offset from the main line by four to eight orders of magnitude (Table III), because the

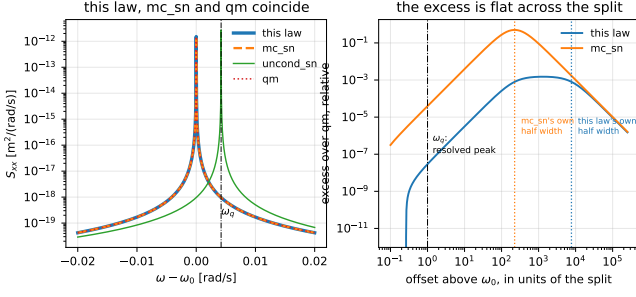


FIG. 4. What the detector sees on the osmium platform: this law’s prediction, the measurement-conditioned alternative’s, and standard quantum mechanics’ main resonances coincide; only unconditional Schrödinger–Newton’s is shifted (left). The excess over standard quantum mechanics at the offset where a resolved second peak would sit, relative to each law’s own linewidth there (right): the excess is present but spread far wider than the offset itself.

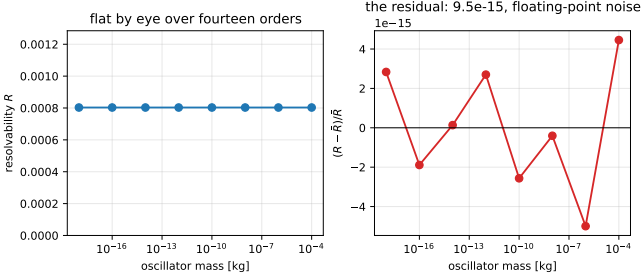


FIG. 5. Resolvability as a function of oscillator mass, at fixed  $\omega_0$ ,  $Q$  and  $T$ : flat to  $9.451e - 15$  (floating-point noise) over fourteen orders of magnitude in mass. A heavier oscillator does not help.

environment’s own records simultaneously suppress the predicted signal’s *weight* and set the resonance’s *width*, through the same localization rate — visible directly in Fig. 4. The obstacle is the line shape, not sensitivity: no integration time resolves a feature far broader than its own offset.

The closed form underlying this, valid in the weak-measurement, small-shift regime (Appendix E),

$$R \equiv \frac{\text{offset}}{\text{width}} = \frac{Q \hbar \omega_{\text{SN}}^2}{8 \omega_0 k_B T}, \quad (7)$$

does not depend on the oscillator’s mass, which the full filtering model confirms to  $9.451e - 15$  over fourteen orders of magnitude (Fig. 5). A larger mass therefore does not help.  $R$  improves with  $Q/T$ : osmium needs  $Q/T \sim 1.246e + 10 \text{ K}^{-1}$  at a millihertz to reach  $R = 1$ , or  $1.246e + 14 \text{ K}^{-1}$  at ten hertz.

A region meeting all three feasibility conditions (resolvability of one, and separation from each alternative within a month) exists: osmium at 1 mHz,  $Q = 10^{12}$  and 1 mK gives  $R = 820.2$ . It is a window rather than a threshold (Fig. 6): far enough above the floor, the resonance becomes too narrow for a periodogram to gather

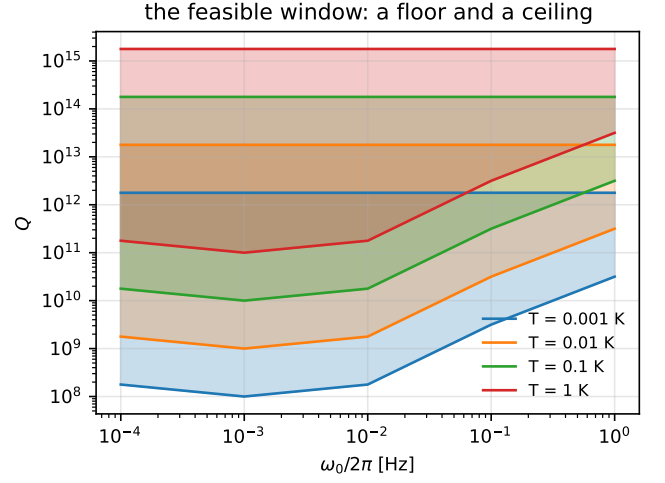


FIG. 6. The feasible region in  $Q$  and  $\omega_0$ , at four temperatures: a *window*, not a threshold. Below the lower curve the second resonance is an unresolved shoulder; above the upper curve, it is resolved but too narrow for a periodogram to gather enough independent samples within a month. The window closes from both sides as  $\omega_0$  rises.

enough independent samples in a month, so raising  $Q$  eventually hurts.

#### D. The observable

Modulating an ancilla’s coupling and watching the raw spectrum does not isolate the effect: the power spectral density at the probe frequency moves by up to  $999.0\times$  its baseline under standard quantum mechanics, which has no self-gravity term, because the ancilla’s backaction heats the oscillator under every law. The second resonance’s *weight*, separated from this heating, passes every control. A protocol of this kind must use that heating-nulled observable.

### VI. WHY THE LIMIT IS THIS LAW’S

The measurement-conditioned alternative does not face the same wall. Its second resonance’s half-width is set by  $\eta_{\text{det}} \Lambda_{\text{ro}}$ , the readout’s own monitoring rate, which the experimenter sets and so can tune down to the imprecision floor,  $\Gamma_2 = 2\gamma$ . This law’s resonance width is set instead by  $\Lambda_{\text{tot}}$ , the environment’s own total record rate, which nobody chooses, giving  $\Gamma_2 = 2\gamma(2\bar{n} + 1)$  — larger by the thermal occupation factor,  $4.167e + 8$  on the osmium platform and  $4.696e + 15$  on the torsion balance. Scanning the readout strength over fifty-six orders of magnitude in the full model, the measurement-conditioned law clears the imprecision floor on the osmium platform with resolvability up to  $4.8 \times 10^4$ ; the present law clears it on no platform.

The law is hard to test not because its signal is small, but because it is sourced by records that no one can choose not to make.

## VII. A MINIMUM OBSERVATION TIME FOR THE WHOLE PROGRAMME

A spectral feature of half-width  $\Gamma$  cannot be resolved in less than about its own coherence time,  $2\pi/\Gamma$ . Every law with a conditioned second resonance — this one and the measurement-conditioned alternative alike — has  $\Gamma_2 \geq 2\gamma$ , forced by the imprecision floor every readout carries (Sec. VI). Combining the two gives a bound independent of which law, platform, or confidence level is in question:

$$t_{\min} = \frac{\pi Q}{\omega_0}, \quad (8)$$

growing *with*  $Q$  — this is the reason the window in Fig. 6 has a ceiling. For a published torsion-balance experiment [12] ( $Q \approx 5 \times 10^4$  at 0.6 mHz), Eq. (8) gives 482.3 days from its quality factor and resonance frequency alone; the experiment reports needing more than 300 days to reach steady state, in agreement in order of magnitude.

The relation underlying Eq. (8), that resolving a spectral feature takes a time set by its inverse width, is standard. Helou *et al.* [4] derive a sharper version for measurement-conditioned Schrödinger–Newton gravity,  $\tau_{\min}(h) \approx 27 h^{-0.73} \times 2/\gamma$  for a Lorentzian peak of normalized height  $h$  at ten per cent confidence (their Eq. 120). Equation (8) is the weaker bound obtained by dropping the dependence on confidence level and signal strength; in exchange, it applies to any law with a conditioned second resonance. I note its generality and its agreement with the run time reported in Ref. [12].

## VIII. RELATION TO OTHER PROPOSALS

*a. Unconditional Schrödinger–Newton.* This law reduces to it exactly with no record ( $f(1) = 1$ ), and departs from it wherever any record exists. It remains unexcluded: the most sensitive published test [12] reports no evidence for a semiclassical-gravity effect and explicitly does not claim to exclude the equation.

*b. Measurement-conditioned Schrödinger–Newton.* Sourced by what an experimenter’s readout has resolved rather than by records that exist [3–5]. It differs from the present law for unread records, and it avoids the resolvability limit of Sec. VI.

*c. Classical-channel and spontaneous-localization models.* Kafri, Taylor and Milburn [6] treat gravity between two resonators as a genuinely stochastic classical measurement channel: like this law, it cannot entangle two masses through gravity alone, but unlike this law, it must add decoherence as the price of not signalling, since a classical channel cannot transmit

quantum correlations noiselessly. Tilloy and Diósi [13] instead source gravity from the state a spontaneous-localization model has already localized, eliminating the single-particle self-interaction and predicting an added decoherence term tied to the localization model’s own parameters. Both share the present law’s premise, that gravity is sourced by what is localized rather than by the full superposition, but replace conditioning on existing records with an intrinsically stochastic mechanism.

*d. Classical–quantum gravity.* Oppenheim [14] couples a genuinely classical spacetime to quantum matter through a trace-preserving, completely positive master equation, forcing quantum mechanics itself to become fundamentally stochastic. This modifies the matter dynamics directly, a different scope than the present law, which modifies only the source term.

*e. Collapse models.* Continuous spontaneous localization at Adler’s parameters [15] already heats a representative platform’s oscillator eight to eleven orders of magnitude faster than its own thermal diffusion rate, excluding it on thermal grounds alone, before any self-gravity question is asked. The Diósi–Penrose criterion [16, 17] and Ghirardi–Rimini–Weber [18] are, like continuous spontaneous localization, collapse mechanisms rather than sourcing laws: they remove a superposition rather than conditioning a source on its records.

*f. Gravitationally-induced entanglement.* Bose *et al.* [7] and Marletto and Vedral [8] propose witnessing the gravitational field’s quantum nature by whether two masses in spatial superposition become entangled through their mutual attraction. This is precisely the test P1 (Sec. III) speaks to: under this law, with no record taken, they would not.

*g. The case for quantizing gravity.* Eppley and Hannah [1] argue on general grounds that a classical field coupled to quantum matter must violate momentum conservation, the uncertainty principle, or no-signalling; Page and Geilker [2] give the experimental half of the case, a Cavendish-type experiment whose source mass’s position is entangled with a radioactive decay, which always deflects toward one branch or the other rather than the expectation value unconditional semiclassical gravity predicts. Both bear on sourcing gravity from the unconditioned mean, not on conditioning it on records.

## IX. LIMITS

1. The selection between the two closed forms of  $f$  rests on one criterion — multiplicative composition — not on a deeper principle. No signalling, in particular, does not distinguish them.
2. The crystal-regime self-gravity frequency  $\omega_{\text{SN}}$ , and the assumption that it enters the oscillator’s frequency squared linearly in the law’s own weight, are imported from the optomechanics literature, not derived from this law.

3. The adapter’s reduction to a thermal oscillator’s state (rank above two) is an extrapolation past where the law is stated for two branches. An independent reading, at the material’s lattice scale rather than through the state’s purity, gives the same conclusion on both platforms where it matters, but the extension itself is not part of the law.
4. The two-body entanglement result (P1) is mean-field, not a full continuous master equation with decoherence and backaction acting throughout.
5. Every decoherence channel in the adapter is treated as resolving position exclusively; a bath that instead recorded momentum would need a different adapter, not built here.
6. Classical force noise — Casimir forces, patch potentials — is not included in the realistic-conditions model. It would add to the total diffusion rate and so strengthen the negative feasibility conclusion, but it has not been shown to be negligible.
7. Two of the three platforms (the torsion balance and the levitated sphere) carry an assumed mechanical quality factor, swept broadly in Sec. V but quoted as single numbers in Table III.
8. The minimum-observation-time bound (Sec. VII) is a special case of a known result [4] and is compared with a single experiment.

## X. CONCLUSION

Gravity sourced by the records that physically exist of a quantum system’s position — rather than by its full wavefunction or by what an observer has measured — is a law that does not signal, by construction, and that makes two sharp predictions: no gravitational entanglement between unrecorded masses, with no added noise; and no self-gravity frequency shift for any oscillator above its motional ground state. Exact simulation with controls confirms both. The signature that distinguishes the law from its measurement-conditioned alternative is out of reach on every platform considered: the records that suppress the signal also broaden the feature that would carry it, independent of the oscillator’s mass. Prediction P2 can be tested with existing apparatus, since none of the platforms in Table III is near its motional ground state, and it separates the law from unconditional Schrödinger–Newton gravity. Finally, the bound of Sec. VII applies to every proposal of this kind: sharpening the resonance to see a conditioned self-gravity signal lengthens the required observation time in proportion.

*a. Data and code availability.* Every number, table and figure in this paper is generated by the `record-conditioned-gravity` repository [19], where

`make all` reproduces them from a clean checkout in under an hour. The repository’s `CLAIMS.md` maps each result to the script and output that produce it.

*b. Origin.* The law arose in work on a separate, unpublished theory of relational quantum gravity. It is stated here independently, and nothing in this paper relies on that theory.

## Appendix A: The closed forms

Equation (5)’s two forms follow directly from the two interpolation rules for  $n_r$  given in Sec. II, by evaluating the weight each places on the non-occupied branch and normalizing to the no-record value; both are closed forms in  $\eta$  alone, implemented as `rcg.law.f_rule_i` and `rcg.law.f_rule_ii`.

## Appendix B: No signalling

**Claim.** For any  $f$  depending only on the reduced state  $\rho$  of the system under study, no local operation on a register entangled with that system can change the law’s prediction.

**Proof sketch.** The reduced state is  $\rho = \text{Tr}_R |\Psi\rangle\langle\Psi|$ , tracing out the register  $R$ . For any unitary  $U_R$  acting on  $R$  alone,  $(\mathbb{I} \otimes U_R)|\Psi\rangle\langle\Psi|(\mathbb{I} \otimes U_R)^\dagger$  has the identical reduced state on the system, since the partial trace is invariant under a unitary on the traced-out factor. Hence  $\eta$ , built from  $\rho$  alone (Eq. (2)), and any  $f(\eta)$ , is unchanged. This includes reading the register (a unitary copying its outcome to a classical pointer, itself part of  $R$ ) and choosing a measurement basis (a unitary rotation of  $R$  before projection); both are covered by the argument and checked numerically in Sec. IV. Two of the alternatives compared are not of this form: sourcing by the measured outcome depends on which basis the register is measured in (a non-unitary projection, not covered by the argument above), and sourcing by how many independent copies of a record exist depends on a count that a unitary redistributing information among those copies can change.

## Appendix C: The adapter

### 1. The filtering equations

For a damped harmonic oscillator under continuous position monitoring at rate  $\mu$  and total momentum diffusion  $D$ , the steady-state covariance  $(V_x, V_{xp}, V_p)$  solves

$$\dot{V}_x = \frac{2}{m} V_{xp} - 8\mu V_x^2, \quad (\text{C1})$$

$$\dot{V}_{xp} = \frac{1}{m} V_p - m\omega_0^2 V_x - \gamma V_{xp} - 8\mu V_x V_{xp}, \quad (\text{C2})$$

$$\dot{V}_p = -2m\omega_0^2 V_{xp} - 2\gamma V_p + 2D - 8\mu V_{xp}^2. \quad (\text{C3})$$

Setting the right-hand sides to zero leaves one quartic in  $V_x$  with positive coefficients, so the positive root is unique. It is found in  $\log V_x$ , since the root can lie twenty orders of magnitude below its  $\mu = 0$  value.  $\mu = D/\hbar^2$  (every channel conditioned on, at unit efficiency) gives the fully conditioned covariance;  $\mu = \eta_{\text{det}}\Lambda_{\text{ro}}$  (the read-out alone) gives the measurement-conditioned one;  $\mu = 0$  gives the unconditioned one. The unconditioned covariance reproduces equipartition, and the fully conditioned one the ground state.

## 2. The extrapolation, checked a second way

The conclusion  $f = 0$  for a thermal oscillator follows from reading  $\eta_{\text{eff}}$  off the unconditioned state's purity, an extension of Eq. (2) beyond the two-branch states for which it is derived. An independent reading instead asks for the reduced state's coherence at the material's own lattice localization length  $\sigma$ ,

$$\eta_{\text{coh}}(\sigma) = \exp\left[-\frac{(\Sigma - \hbar^2/4V_x)\sigma^2}{2\hbar^2}\right], \quad \Sigma \equiv V_p - \frac{V_{xp}^2}{V_x}, \quad (\text{C4})$$

which shares no algebra with the purity-based route. On both hot platforms (osmium, torsion balance) the thermal de Broglie length is  $10^{-16}$  m or smaller against a lattice localization of a few picometres, so  $\eta_{\text{coh}}(\sigma) \rightarrow 0$ , and the conclusion  $f = 0$  does not depend on which reading is used.

## Appendix D: Platform parameters

### Appendix E: The closed-form resolvability, and its regime of validity

Equation (7) assumes two things: weak measurement ( $4m\Lambda\hbar \ll (m\omega_0)^2$ ), so the fully conditioned variance is the ground state's, and a small shift ( $\omega_{\text{SN}} \ll \omega_0$ , so the offset is  $\omega_{\text{SN}}^2/2\omega_0$ ). Checked against the full filtering model, it is accurate to within a few per cent on the osmium platform and within tens of per cent on the ground-state-cooled sphere, but wrong by six orders of magnitude on the torsion balance, which is deep in the strong-measurement regime and has  $\omega_{\text{SN}}$  above  $\omega_0$ . All quoted numbers come from the full filtering model; Eq. (7) is used only for scaling arguments, where it holds.

- 
- [1] K. Eppley and E. Hannah, *Foundations of Physics* **7**, 51 (1977).
  - [2] D. N. Page and C. D. Geilker, *Physical Review Letters* **47**, 979 (1981).
  - [3] H. Yang, H. Miao, D.-S. Lee, B. Helou, and Y. Chen, *Physical Review Letters* **110**, 170401 (2013), [arXiv:1210.0457](#).
  - [4] B. Helou, J. Luo, H.-C. Yeh, C. gang Shao, B. J. J. Slagmolen, D. E. McClelland, and Y. Chen, *Physical Review D* **96**, 044008 (2017), [arXiv:1612.06310](#).
  - [5] Y. Liu, H. Miao, Y. Chen, and Y. Ma, *Physical Review D* **107**, 024004 (2023), [arXiv:2207.05966](#).
  - [6] D. Kafri, J. M. Taylor, and G. J. Milburn, *New Journal of Physics* **16**, 065020 (2014), [arXiv:1401.0946](#).
  - [7] S. Bose, A. Mazumdar, G. W. Morley, H. Ulbricht, M. Toroš, M. Paternostro, A. A. Geraci, P. F. Barker, M. S. Kim, and G. Milburn, *Physical Review Letters* **119**, 240401 (2017), [arXiv:1707.06050](#).
  - [8] C. Marletto and V. Vedral, *Physical Review Letters* **119**, 240402 (2017), [arXiv:1707.06036](#).
  - [9] A. Großardt, J. Bateman, H. Ulbricht, and A. Bassi, *Physical Review D* **93**, 096003 (2016), [arXiv:1510.01696](#).
  - [10] T. Westphal, H. Hepach, J. Pfaff, and M. Aspelmeyer, *Nature* **591**, 225 (2021), [arXiv:2009.09546](#).
  - [11] U. Delić, M. Reisenbauer, K. Dare, D. Grass, V. Vuletić, N. Kiesel, and M. Aspelmeyer, *Science* **367**, 892 (2020).
  - [12] T. Yan, Y. Liu, L. Prokhorov, J. Smetana, H. Miao, Y. Ma, V. Boyer, and D. Martynov, *Physical Review D* **111**, 082007 (2025), [arXiv:2411.17817](#).
  - [13] A. Tilloy and L. Diósi, *Physical Review D* **93**, 024026 (2016), [arXiv:1509.08705](#).
  - [14] J. Oppenheim, *Physical Review X* **13**, 041040 (2023), [arXiv:1811.03116](#).
  - [15] A. Bassi, K. Lochan, S. Satin, T. P. Singh, and H. Ulbricht, *Reviews of Modern Physics* **85**, 471 (2013).
  - [16] L. Diósi, *Physical Review A* **40**, 1165 (1989).
  - [17] R. Penrose, *General Relativity and Gravitation* **28**, 581 (1996).
  - [18] G. Ghirardi, A. Rimini, and T. Weber, *Physical Review D* **34**, 470 (1986).
  - [19] D. T. Swanson, record-conditioned-gravity: a sourcing law, its computational check, and its testability under realistic conditions (2026).

TABLE IV. All three platforms' parameters, with sources and assumptions (expanding Table III).

| platform     | $m$ [kg]  | $\omega_0/2\pi$ [Hz] | $T$ [K]   | $Q$       | notes (citation:<br>data/platforms.yaml)                                           |
|--------------|-----------|----------------------|-----------|-----------|------------------------------------------------------------------------------------|
| osmium-paul  | 1.661e-12 | 1.000e+01            | 1.000e-01 | 1.664e+09 | every number [G16] except the residual gas species ([9])                           |
| torsion-mg   | 9.000e-05 | 2.600e-03            | 2.930e+02 | 5.000e+00 | $Q = 5$ is ASSUMED; swept in stage 3 ([10])                                        |
| levitated-gs | 3.368e-18 | 3.050e+05            | 1.218e-05 | 1.000e+08 | silica taken as silicon for sigma, which OVERSTATES w_SN; $Q = 1e8$ ASSUMED ([11]) |